Essential in SOLIDWORKS (40 Hours)

S.No	Session	Topics	Type	Duration (Hrs)
1	Session 1	Introduction to CAD, CAE, PDM	Theory/Demo	2 Hrs
		Introduction to Solidworks		
		Features of SolidWorks		
		Various products available in SolidWorks for Product Design, Simulation, Communication		
		SolidWorks Graphical User Interface: Feature manager design tree, Callouts, Handles, Confirmation corner, mouse buttons, keyboard shortcuts, Command Manager,		
		Hardware and Software requirements, SolidWorks Task Scheduler, SolidWorks Rx		
2	Session 2	Exercises included for practice	Practicals	2 Hrs
3	Session 3	**Sketch Entities:** Inference line, Centerline line, Line, Circle, Arc, Ellipse, Rectangle, Slots, Polygon, Parabola, Ellipse, Partial Ellipse, Spline, Spline tools, Spline on surface, Equation driven curve, Points, Text, Construction geometry, Snap, grid	Theory/Demo	2 Hrs
4	Session 4	Exercises included for practice	Practicals	2 Hrs
5	Session 5	**Sketch Tools:** Fillet, Chamfer, Offset, Convert entities, Trim, Extend, Split, Jog, Mirror, Dynamic Mirror, Move, Copy, Rotate, Scale, Stretch, Sketch pattern	Theory/Demo	2 Hrs
		Make path, Close Sketch To Model, Sketch picture, Check Sketch for Feature, Area hatch/Fill		
		Blocks: Make block, Edit block, Insert block, Add/Remove Entities, Rebuild, Save, Explode		
		Relations: Adding Sketch Relation, Automatic relations		
		Dimensioning: Smart, Horizontal, Vertical, Ordinate, Horizontal ordinate, Vertical ordinate, Align ordinate, Fully define sketch		
		Sketch Diagnosis, SketchXpert, 3D Sketching, Rapid Sketch		
6	Session 6	Exercises included for practice	Practicals	2 Hrs
7	Session 7	Part Modeling Tools	Theory/Demo	2 Hrs
		Creating reference planes		
		Creating Extrude features: Direction1, Direction2, From option, Thin feature, Applying draft, Selecting contours		
		Creating Revolve features: Selecting Axis, Thin features, Selecting contours		
		Creating Swept features: Selecting, Profile and Path, Orientation/twist type, Path Alignment, Guide Curves, Start/End tangency, Thin feature		
8	Session 8	Exercises included for practice	Practicals	2 Hrs
9	Session 9	**Creating Loft features:** Selecting Profiles, Guide curves, Start/End Constraints, Centerline parameters, Sketch tools, Close loft.	Theory/Demo	2 Hrs
		Selecting geometries: Selection Manager, Multiple Body concepts		
		Creating Reference: Points, axis, coordinates		
10	Session 10	Exercises included for practice	Practicals	2 Hrs
11	Session 11	Creating curves	Theory/Demo	2 Hrs
		Split curve, Project curve, Composite curve, Curve through points, Helix and Spiral		
		Creating Fillet features		
		Inserting Hole types		
12	Session 12	Exercises included for practice	Practicals	2 Hrs
13	Session 13	Creating Chamfer	Theory/Demo	2 Hrs
		Creating Shell, Creating Rib		
		Creating Pattern: Linear pattern, Circular pattern, Sketch driven pattern, Curve driven pattern, Table driven pattern, Fill pattern, mirro		
14	Session 14	Exercises included for practice	Practicals	2 Hrs
15	Session 15	Assembly Modeling Tools	Theory/Demo	2 Hrs
		Introduction to Assembly Modeling & Approaches: Top down and Bottom up approach		
		Applying Standard Mates: Coincident, Parallel, Perpendicular, Tangent, Concentric, Lock, Distance, Angle		
16	Session 16	Exercises included for practice	Practicals	2 Hrs
17	Session 17	Generating Drawing Views	Theory/Demo	2 Hrs
		Introduction To Angle Of Projection		
		Generating Views: Generating Model View, Projected Views, Inserting Standard 3 View		
		View creation relative to model, Inserting predefined views, empty views, Auxiliary Views, Detailed Views, Crop view, Broken –Out Section, Broken Views, Section View, Aligned Section View, Alternate Position View, Working assembly specific view, Drawing properties, Manipulating views		

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18	Session 18	Exercises included for practice	Practicals	2 Hrs
19	Session 19	Sheet Metal Design	Theory/Demo	2 Hrs
		Concepts in Sheet metal design bend allowance bend deduction, K-factor		
		Inserting Base Flange, Sheet Metal Tab, Edge Flange, Miter Flange, Hem, Jog		
		Creating Break Corner/Corner Trim, Closed Corners, Rip		
		Inserting Sketched Bend, Fold/Unfold, Forming Tools		
20	Session 20	Exercises included for practice	Practicals	2 Hrs